

**Syllabus:** Introduction to Fourier transform, Introduction to communication system, Need for modulation, Frequency Division Multiplexing, Amplitude Modulation, Time domain and Frequency domain descriptions, Single tone modulation, Power relations in AM waves, Generation of AM waves: Square law Modulator, Switching modulator, Detection of AM Waves: Square law detector, Envelope detector, Related problems.

### Basic Key terms

- \* **Communication:** It is the process whereby information is transferred from one point (source) to the other point (destination). So simply communication is exchanging of information in between two points/persons/places.
- \* **Information:** It is the news one wishes to convey. Simply it is a data.
- \* **Message:** Message is the physical manifestation of information as produced by the source.
- \* **Signal:** It is a physical quantity which carries some information and varies with time, space or any independent variable.
- \* **Signal types:** Basically signals are classified under two categories as Analog and Discrete time signals.
  - **Analog-** Continuous function of time with continuous amplitude.
  - **Discrete time-** Discrete function of time with continuous amplitude.
  - **Digital signal-** Quantized and coded signal (Advance of Discrete time).

### Introduction to Fourier transform

Fourier Transform is a Transformation Technique which transforms signals from the continuous-time domain into Frequency domain and vice-versa. The limitation of Fourier Series is that, it is applicable only to periodic signals. It can't be applicable to Aperiodic signals. The Fourier Transform overcomes this limitation because it is applicable to both aperiodic and also Periodic signals.

Fourier Transform can be developed by finding the Exponential Fourier series of a periodic signal and then tending 'T' to infinity.

Fourier Transform is extremely useful mathematical tool and is extensively used in the analysis of LTI systems, cryptography, signal analysis, signal processing, astronomy etc.

Note: Fourier Transform does not exist for all the Aperiodic signals. The conditions for the existence of Fourier Transform should be satisfied which are called as Dirichlet conditions.

## FOURIER TRANSFORM OF A NON-PERIODIC SIGNAL FROM THE FOURIER SERIES OF PERIODIC SIGNALS

Let  $x(t)$  is a Non Periodic signal and  $x_T(t)$  is a periodic with period  $T$ . Then relation is given by

$$x(t) = \lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad \text{--- (1)}$$

We know Exponential Fourier Series of periodic signal  $x_T(t)$  is

$$x_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$\text{where } C_n = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-jn\omega_0 t} dt \quad \text{and}$$

$$\omega_0 = \frac{2\pi}{T}$$

$C_n$  represents amplitude of component of frequency  $n\omega_0$ .

$$\text{So, } T \cdot C_n = \int_{-T/2}^{T/2} x_T(t) e^{-jn\omega_0 t} dt \quad \text{--- (2)}$$

Let  $n\omega_0 = \omega$  at  $T \rightarrow \infty$ .

$\Rightarrow \omega_0 = \frac{2\pi}{T} \rightarrow 0$ , then discrete spectrum becomes continuous. Then summation becomes integral and  $x_T(t) \rightarrow x(t)$ . Then from Eq (2)

$$T C_n = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x_T(t) e^{-jn\omega_0 t} dt$$

$$= \int_{-\infty}^{\infty} \left[ \lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \right] e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \quad [\because \text{from Eq (1)}]$$

$$T C_n = X(\omega) \quad \text{--- (3)}$$

$$\therefore \text{Fourier Spectrum of } x(t) \text{ is } \boxed{X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt} \quad \text{--- (4)}$$

$$\text{we have } x_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{\infty} \frac{X(\omega)}{T} \cdot e^{jn\omega_0 t} \quad [\because C_n = \frac{X(\omega)}{T}]$$

$$= \sum_{n=-\infty}^{\infty} \frac{X(\omega)}{2\pi} \cdot e^{jn\omega_0 t} \cdot \omega_0 \quad [\because T = \frac{2\pi}{\omega_0}]$$

$$\Rightarrow x(t) = \lim_{T \rightarrow \infty} x_T(t) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(\omega) \cdot e^{jn\omega_0 t} \cdot \omega_0$$

As  $T \rightarrow \infty$ ,  $\omega_0 = \frac{2\pi}{T} \rightarrow 0$  (Infinitely Small) That change can be represented as  $d\omega$

$$\therefore x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega \quad \text{--- (5)}$$

Eq (5) is called as Inverse Fourier Transform of  $X(\omega)$ .

$X(\omega)$  and  $x(t)$  are known as Fourier Transform pair.

$$X(\omega) = FT[x(t)] \quad \text{and} \quad x(t) = FT^{-1}[X(\omega)]$$

### MAGNITUDE AND PHASE REPRESENTATION OF FOURIER TRANSFORM.

The Magnitude and phase representation of the Fourier Transform is a tool used to Analyse the Transformed signal.

Let  $x(\omega)$  is a complex function of  $\omega$ .

$$\text{i.e. } x(\omega) = x_R(\omega) + j x_i(\omega)$$

Then Magnitude is given by  $|x(\omega)| = \sqrt{x_R(\omega)^2 + x_i(\omega)^2}$

$$\text{phase of } x(\omega) \text{ is given by } \angle x(\omega) = \tan^{-1} \left( \frac{x_i(\omega)}{x_R(\omega)} \right)$$

Note: \* The plot of  $|x(\omega)|$  versus  $\omega$  is known as amplitude spectrum.

\* The plot of  $\angle x(\omega)$  versus  $\omega$  is known as phase spectrum.

\* Amplitude and phase ~~together~~ spectra together called Frequency spectrum.

### EXISTENCE OF FOURIER TRANSFORMS :

The Fourier Transform does not exist for all Aperiodic functions. The conditions for a function  $x(t)$  to have Fourier Transform, called Dirichlet's conditions are:

1.  $x(t)$  is absolutely integrable over the interval  $-\infty$  to  $\infty$ , That is  $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
2.  $x(t)$  has a finite number of discontinuities in every finite time interval. Further each of these discontinuities must be finite.
3.  $x(t)$  has a finite number of maxima and minima in every finite time interval.

Note: Dirichlet conditions are sufficient conditions But not necessary condition.



## FOURIER TRANSFORMS OF STANDARD SIGNALS:

### (a) Impulse Function $\delta(t)$ :-

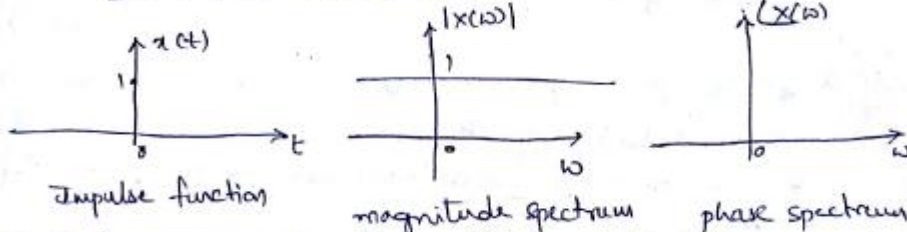
$$\text{given } x(t) = \delta(t), \quad \delta(t) = \begin{cases} 1 & \text{for } t=0 \\ 0 & \text{for } t \neq 0 \end{cases}$$

$$\begin{aligned} \text{Then } X(\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt \\ &= e^{-j\omega t} \Big|_{t=0} = 1 \end{aligned}$$

$$\therefore X(\omega) = F[\delta(t)] = 1$$

$$\Rightarrow |X(\omega)| = 1 \text{ for all } \omega$$

$$\angle X(\omega) = 0 \text{ for all } \omega$$



$$\text{Similarly, } F[\delta(t-t_0)] = \int_{-\infty}^{\infty} \delta(t-t_0) e^{-j\omega t} dt = e^{-j\omega t_0}$$

### (c) Complex exponential function ( $e^{j\omega_0 t}$ ) :-

To find Fourier Transformation of  $e^{j\omega_0 t}$ , Consider finding the Inverse Fourier Transformation of  $\delta(\omega-\omega_0)$ .

$$\text{let } X(\omega) = \delta(\omega-\omega_0)$$

$$\begin{aligned} x(t) &= F^{-1}[X(\omega)] = F^{-1}[\delta(\omega-\omega_0)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega-\omega_0) \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t} \end{aligned}$$

$$\therefore F^{-1}[\delta(\omega-\omega_0)] = x(t) = \frac{1}{2\pi} e^{j\omega_0 t}$$

$$\Rightarrow x(t) = F^{-1}[2\pi \delta(\omega-\omega_0)] = e^{j\omega_0 t}$$

$$\Rightarrow F[x(t)] = F[e^{j\omega_0 t}] = 2\pi \delta(\omega-\omega_0)$$

### (d) Cosine wave ( $\cos \omega_0 t$ ) :

$$\text{given } x(t) = \cos \omega_0 t$$

$$X(\omega) = F[x(t)] = \int_{-\infty}^{\infty} \cos \omega_0 t \cdot e^{-j\omega t} dt$$

$$= F\left[\frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]\right]$$

$$= \frac{1}{2} [F(e^{j\omega_0 t}) + F(e^{-j\omega_0 t})]$$

$$= \frac{1}{2} [2\pi \delta(\omega-\omega_0) + 2\pi \delta(\omega+\omega_0)]$$

$$X(\omega) = \pi [\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$$

## FOURIER TRANSFORM OF A PERIODIC SIGNAL :

Fourier Transform can be used as a universal mathematical tool in the Analysis of both non-periodic and periodic waveform over the entire interval. Fourier Transform of Periodic functions may be found using Concept of impulse function.

Let a signal  $x_T(t)$  is a periodic signal with time  $T$ . then  $x_T(t)$  is represented in terms exponential Fourier series as

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad \text{--- (1)}$$

Applying Fourier Transformation to Equation (1)

$$\begin{aligned} X(\omega) &= F[x(t)] = F\left[\sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}\right] \\ &= \sum_{n=-\infty}^{\infty} C_n F[e^{jn\omega_0 t}] \quad \text{--- (2)} \end{aligned}$$

using frequency shifting theorem, we have

$$F[e^{jn\omega_0 t}] = 2\pi \delta(\omega - n\omega_0)$$

then Equation (2) can be written as

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} C_n \cdot 2\pi \delta(\omega - n\omega_0) \\ &= 2\pi \sum_{n=-\infty}^{\infty} C_n \cdot \delta(\omega - n\omega_0) \end{aligned}$$

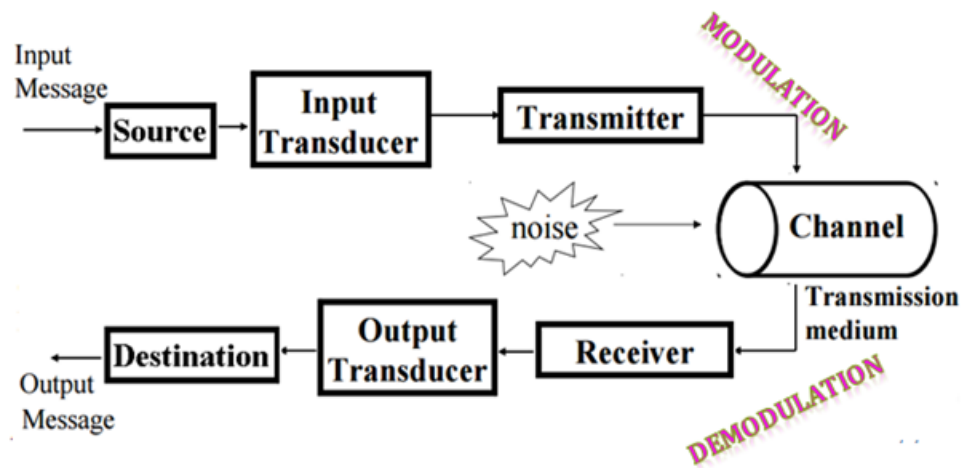
where  $C_n$  - Fourier coefficient associated with  $x(t)$  and is given by  $C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt$ .

Note: The Fourier Transform of periodic function consists of a Train of equally spaced impulses. The strength (area) of each impulse is given by  $2\pi C_n$ .

## Introduction to Communication System

The following are the elements of communication system.

- \* Information Source
- \* Input Transducer
- \* Transmitter
- \* Communication Channel/ Medium
- \* Receiver
- \* Output Transducer
- \* Destination



**Figure 1.1 Elements of Communication Systems**

**Information Source:** Its function is to produce required message which has to be transmitted.

**Input Transducer:** A transducer is a device which can convert one form of energy into another form. The signal from information may or may not be electrical in nature. Input transducer is used to convert non-electrical signal into electrical form.

**Transmitter:** Its functionality is processing of message signal into a form suitable for transmission over the communication channel. A process called modulation is used in transmission section.

**Communication channel:** It is used to provide a pathway between the transmitter's output and the receiver's input. It may be either wired or wireless. For example co-axial cables, optical fibres, free space etc,. When a transmitted signal propagates along the channel it may get distorted due to channel imperfection. Moreover, noise adds to the transmitted signal and the receiver receives a corrupted version of the transmitted signal.

**Receiver:** The job of the receiver is processing of the received signal to recover the appropriate message signal. If the different elements do their jobs accordingly, then the output signal should equal to the input message signal.

**Output Transducer:** A transducer is a device which can convert one form of energy into another form. The signal from receiver is electrical in nature. Output transducer is used to convert electrical signal into original information signal.

**Destination:** The reconstructed original message signal is delivered to the user of information.

## **Introduction to Modulation**

- \* *Modulation* is a process by which some characteristics of carrier signal is varied in accordance with the instantaneous value of Modulating/message signal.
- \* *Demodulation* is the process of recovering original message/baseband signal from Modulated signal

- \* *Modulating/message signal/ Baseband signal is a LOW frequency signal (Audio frequency signal).*
- \* *Carrier signal is a HIGH frequency signal (Radio frequency signal).*
- \* *The resulting signal from the process of Modulation is called Modulated signal.*

**Need for Modulation:** Modulation is required in communication for following reasons.

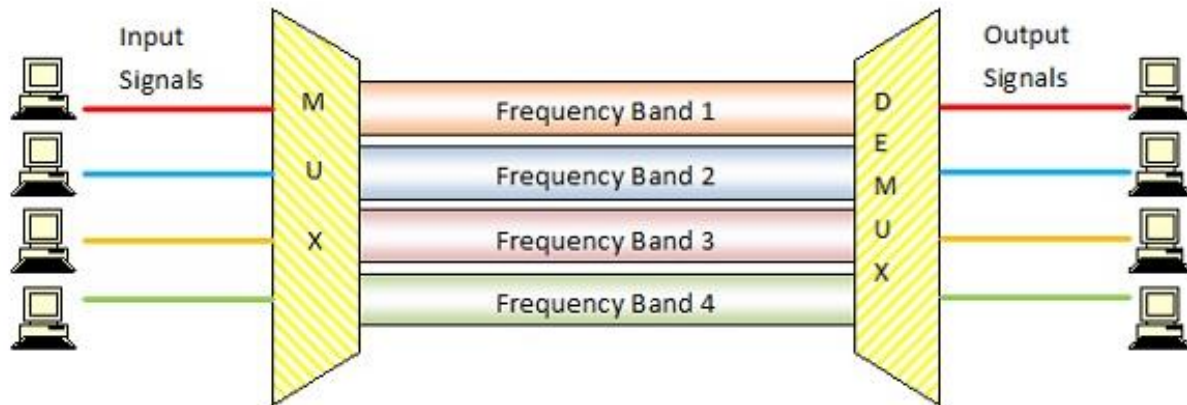
- ✓ ***Efficient Radiation:*** In radio communications information is transmitted in the form of Electromagnetic waves from transmitting antenna has length (L) of  $1/4^{\text{th}}$  of radiated signal wavelength ( $\lambda$ ). ***Example:*** If  $F = 5\text{KHz}$  then  $L = \frac{\lambda}{4} = \frac{c}{4F} = \frac{3 \times 10^8}{4 \times 5 \times 10^3} = 15\text{Km}$  (Practically impossible). If AF signal Transmitted to a RF signal of  $3\text{MHz}$  then  $L = \frac{\lambda}{4} = \frac{c}{4F} = \frac{3 \times 10^8}{4 \times 3 \times 10^6} = 25\text{m}$  (Practically possible). Where 'C' is velocity of Light.
- ✓ ***Frequency Translation:*** Modulation enables one to translate the signals occupying similar frequency ranges to different regions in the frequency spectrum. This allows a user to tune his Radio/TV to a particular broadcasting channel. ***Example:*** Audio signal range  $20\text{Hz} - 20\text{KHz}$ . All the radio stations use same range then programmes of different stations will get mixed up.
- ✓ ***Multiplexing:*** Sometimes it is necessary to send a number of signals simultaneously between two points. Modulation scheme enables one to multiplex multiple signals at same time in a single channel without any interference themselves. Used in Long distance telephony, data telemetry etc.
- ✓ ***Reduction of Noise:*** Noise and Interference are the two major limitations of any communication system. Some modulation schemes can suppress the noise and interference to some extent.

## **Frequency Division Multiplexing**

Frequency division multiplexing (FDM) is a technique of multiplexing which means combining more than one signal over a shared medium. In FDM, signals of different frequencies are combined for concurrent transmission.

In FDM, the total bandwidth is divided to a set of frequency bands that do not overlap. Each of these bands is a carrier of a different signal that is generated and modulated by one of the sending devices. The frequency bands are separated from one another by strips of unused frequencies called the guard bands, to prevent overlapping of signals. The modulated signals are combined together using a multiplexer (MUX) in the sending end. The combined signal is transmitted over the communication channel, thus allowing multiple independent data streams to be transmitted simultaneously. At the receiving end, the individual signals are extracted from the combined signal by the process of demultiplexing (DEMUX).

The following diagram conceptually represents multiplexing using FDM. It has 4 frequency bands, each of which can carry signal from 1 sender to 1 receiver. Each of the 4 senders is allocated a frequency band. The four frequency bands are multiplexed and sent via the communication channel. At the receiving end, a demultiplexer regenerates the original four signals as outputs.



**Figure 1.2 Frequency Division Multiplexing**

Here, if the frequency bands are of 150 KHz bandwidth separated by 10KHz guard bands, then the capacity of the communication channel should be at least 630 KHz (channels :  $150 \times 4$  + guard bands :  $10 \times 3$ ).

#### **Advantages of FDM:**

1. **Efficient Use of Bandwidth:** FDM allows multiple signals to be transmitted over a single communication channel, which can lead to more efficient use of available bandwidth.
2. **No Time Synchronization Required:** FDM does not require precise time synchronization between the transmitting and receiving devices, making it easier to implement.
3. **Low Implementation Cost:** FDM is a relatively simple technique that does not require sophisticated hardware or software, making it less expensive to implement.

#### **Disadvantages of FDM:**

1. **Limited Capacity:** FDM is limited in terms of the number of signals that can be transmitted over a single communication channel, which can be a disadvantage in applications where a large number of signals need to be transmitted.
2. **Interference:** FDM can be susceptible to interference from other signals transmitted on nearby frequencies, which can degrade the quality of the transmitted signals.
3. **Difficulty in Assigning Frequencies:** FDM requires careful assignment of frequencies to different signals to avoid interference, which can be a complex and time-consuming process.



### Application of FDM:

1. In the first generation of mobile phones, FDM was used.
2. The use of FDM in television broadcasting
3. FDM is used to broadcast FM and AM radio frequencies.

### Introduction to Amplitude Modulation

- \* **Amplitude Modulation** is defined as it is the process in which the amplitude of Carrier wave varies in accordance with the instantaneous value of Modulating/Message signal.

### Time domain Analysis:

- \* Let carrier signal  $c(t) = A_c \cos \omega_c t$ , where  $\omega_c = 2\pi f_c$  and message signal is  $m(t) = A_m \cos \omega_m t$ , where  $\omega_m = 2\pi f_m$ .
- \* The Standard AM signal in time domain is  $S_{AM}(t) = A_c[1 + k_a m(t)] \cos \omega_c t$ , where  $k_a$  is Amplitude sensitivity.
- \* Substituting message signal,  $m(t)$  in Standard AM signal Equation, we may get AM wave as  $S_{AM}(t) = A_c[1 + k_a A_m \cos \omega_m t] \cos \omega_c t$ . where  $k_a A_m$  is called **Modulation index/factor-Dimensionless**.
- \* The amplitude modulated (AM) signal consists of both modulated carrier signal and unmodulated carrier signal.
- \* **100% or Percentage Modulation:**  $|k_a m(t)| = k_a A_m = \mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = 1$
- \* **Under Modulation:**  $|k_a m(t)| = k_a A_m = \mu < 1$
- \* **Over Modulation:**  $|k_a m(t)| = k_a A_m = \mu > 1$

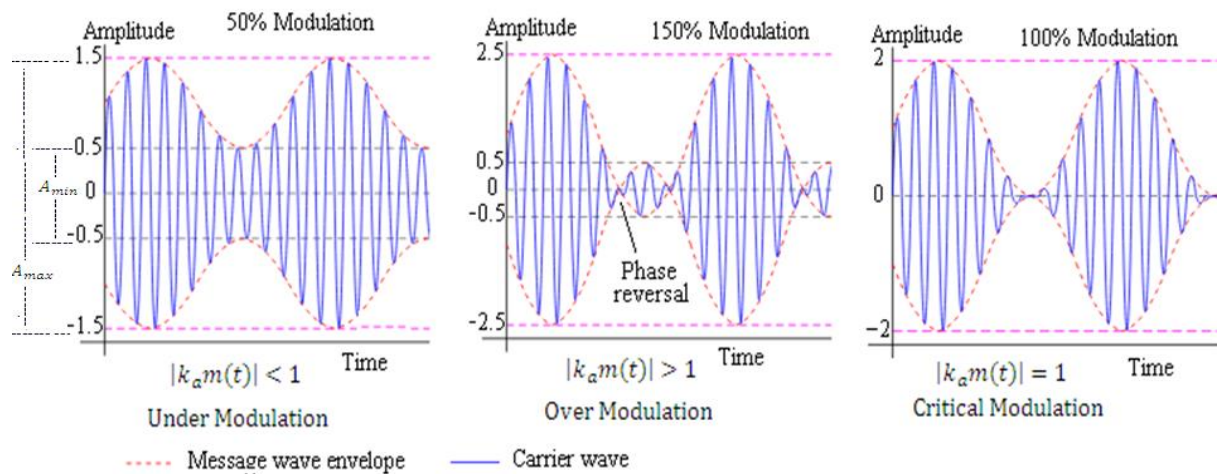


Figure 1.3 Amplitude Modulation

### Frequency domain Analysis:

To represent AM signal in frequency domain we have to apply Fourier transform. We know the Standard AM signal in time domain is represented as follows

$$S_{AM}(t) = A_c[1 + k_a m(t)] \cos \omega_c t \text{ -----(1)}$$

Substituting message signal in Eq(1),  $S_{AM}(t) = A_c[1 + k_a A_m \cos \omega_m t] \cos \omega_c t$  -----(2)

By Simplifying equation(2), we will get

$$S_{AM}(t) = A_c \cos \omega_c t + \frac{k_a A_m A_c}{2} [\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]$$

Since,  $k_a A_m = \mu$ , the above equation is simplified as

$$\Rightarrow S_{AM}(t) = A_c \cos \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t] \quad \text{-----(3)}$$

(i)

(ii)

(iii)

(i)---- Unmodulated Carrier Wave, (ii)---- Upper sideband, (iii)---- Lower sideband

To represent Frequency domain analysis, we have to apply Fourier Transform to Eqs-(3)

$$S_{AM}(\omega) = A_c \pi [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \frac{\mu A_c}{2} \pi [\delta(\omega - (\omega_c + \omega_m)) + \delta(\omega + (\omega_c + \omega_m))] + \frac{\mu A_c}{2} \pi [\delta(\omega - (\omega_c - \omega_m)) + \delta(\omega + (\omega_c - \omega_m))] \quad \text{-----(4)}$$

② Complex exponential function ( $e^{j\omega_0 t}$ ) :-

To find Fourier Transformation of  $e^{j\omega_0 t}$ , Consider finding the inverse Fourier Transformation of  $\delta(\omega - \omega_0)$

$$\text{Let } X(\omega) = \delta(\omega - \omega_0)$$

$$\begin{aligned} x(t) &= F^{-1}[X(\omega)] = F^{-1}[\delta(\omega - \omega_0)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t} \end{aligned}$$

$$\begin{aligned} \therefore F^{-1}[\delta(\omega - \omega_0)] &= x(t) = \frac{1}{2\pi} e^{j\omega_0 t} \\ \Rightarrow x(t) &= F^{-1}[2\pi \delta(\omega - \omega_0)] = e^{j\omega_0 t} \\ \Rightarrow F[x(t)] &= F[e^{j\omega_0 t}] = 2\pi \delta(\omega - \omega_0) \end{aligned}$$

③ Cosine wave ( $\cos \omega_0 t$ ) :-

$$\text{given } x(t) = \cos \omega_0 t$$

$$\begin{aligned} X(\omega) &= F[x(t)] = \int_{-\infty}^{\infty} \cos \omega_0 t \cdot e^{-j\omega t} dt \\ &= F\left[\frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]\right] \\ &= \frac{1}{2} [F(e^{j\omega_0 t}) + F(e^{-j\omega_0 t})] \\ &= \frac{1}{2} [2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)] \\ X(\omega) &= \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \end{aligned}$$

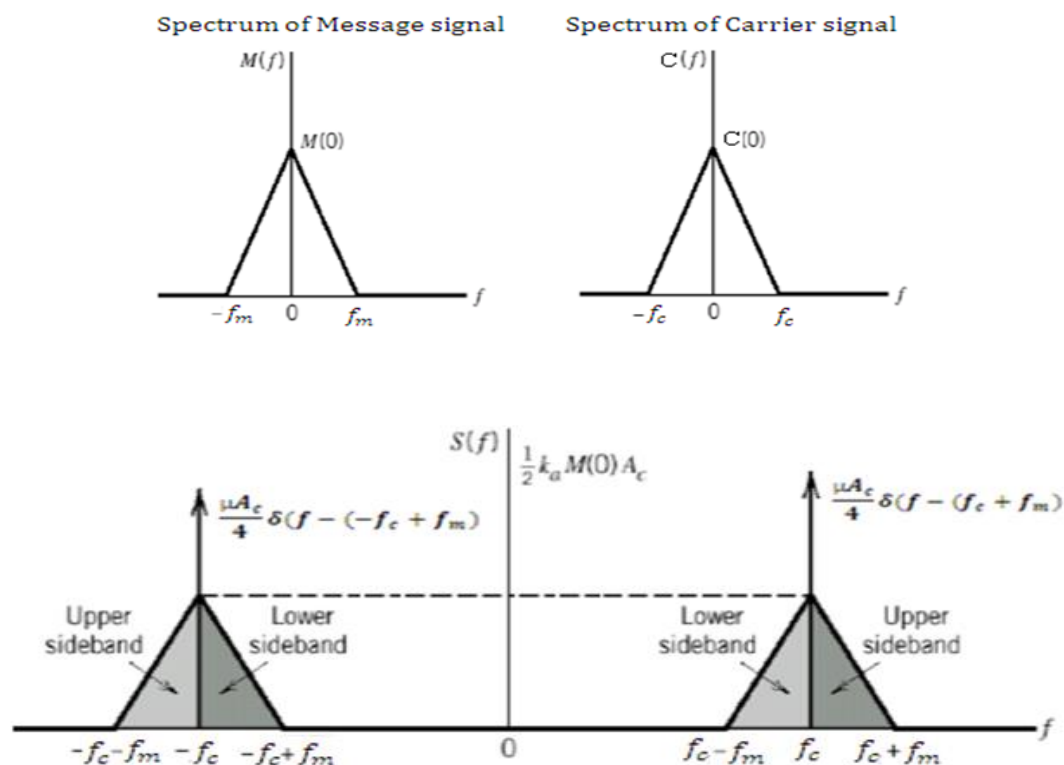


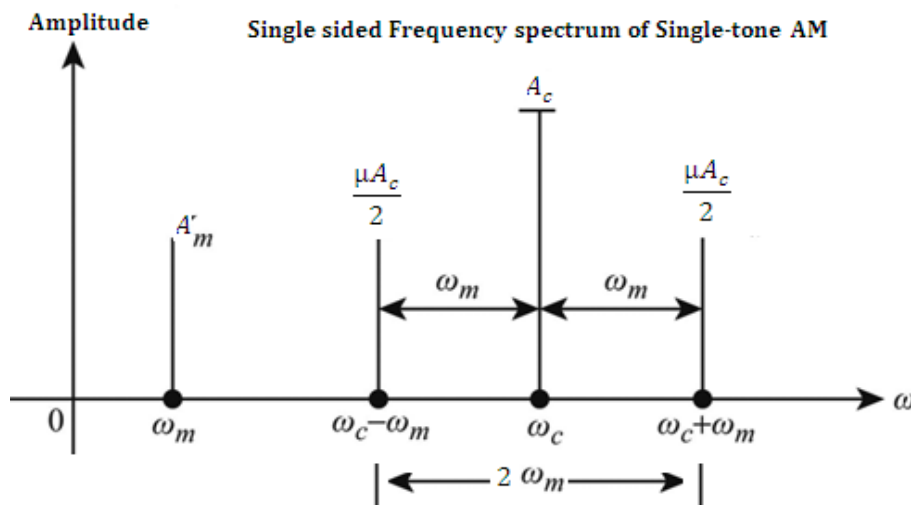
Figure 1.4 Spectrum of AM wave

### ❖ **Single-tone Amplitude Modulation:**

- ✓ It is a process in which modulating signal consists of only a single frequency component.
- ✓ The mathematical expression for Single-tone AM is given by

$$S_{AM}(t) = A_c \cos \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c + \omega_m)t] + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m)t]$$

- ✓ Where  $\mu = k_a A_m = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$  is called **Modulation index/factor** defined as measure of extent of amplitude variation/deviation about an unmodulated carrier wave. It is dimensionless.



**Figure 1.5 Spectrum of AM wave in samples**

**Modulation index for multi-tone Sinusoidal modulation:** Multi-tone modulation is a process in which modulating signal consists more than one frequency component.

A carrier is modulated by a number of Sinusoidal modulating signals of frequencies  $f_1, f_2, f_3, \dots$  etc.

Then total power,  $P_t = \frac{A_c^2}{2} + \frac{\mu_1^2 A_c^2}{4} + \frac{\mu_2^2 A_c^2}{4} + \frac{\mu_3^2 A_c^2}{4} + \dots$

$$P_t = \frac{A_c^2}{2} \left[ 1 + \frac{\mu_t^2}{2} \right]$$

$$\Rightarrow P_t = P_c \left[ 1 + \frac{\mu_t^2}{2} \right]$$

$$\text{where } \mu_t^2 = \mu_1^2 + \mu_2^2 + \mu_3^2 + \dots$$

$$\mu_t = \sqrt{\mu_1^2 + \mu_2^2 + \mu_3^2 + \dots}$$

### **Power calculation in Amplitude Modulation:**

AM wave has three terms as unmodulated carrier, upper sideband and lower sideband. So in AM, the total radiated power is obtained as sum of power consumed by carrier and two sidebands.

$$\begin{aligned} \text{Carrier Power, } P_c &= \text{Mean square value of } A_c \cos \omega_c t \\ &= \overline{[A_c \cos \omega_c t]^2} = \frac{1}{2\pi} \int_0^{2\pi} A_c^2 \cos^2 \omega_c t \cdot d\theta \\ &= \frac{A_c^2}{2} \end{aligned}$$

Sideband power,  $P_s$  = Mean square value of  $m(t) \cdot \cos \omega_c t$

$$\begin{aligned}
 P_s &= \overline{(m(t) \cos \omega_c t)^2} \\
 &= \frac{1}{2\pi} \int_0^{2\pi} m^2(t) \cos^2 \omega_c t \, dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} [2 \cos^2 \omega_c t] m^2(t) \, dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} [1 + \cos 2\omega_c t] m^2(t) \, dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} m^2(t) \, dt + \frac{1}{2\pi} \int_0^{2\pi} \frac{m^2(t)}{2} \cos 2\omega_c t \, dt
 \end{aligned}$$

In AM generation, BPF is used to filter out the second integration term of carrier frequency  $\omega_c$

$$\therefore P_s = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} m^2(t) \, dt = \frac{1}{2} \overline{m^2(t)}$$

$$\begin{aligned}
 P_{LSB} = P_{USB} &= \frac{P_s}{2} = \frac{1}{4} \overline{m^2(t)} \\
 &= \frac{1}{4} (A_m \cos \omega_m t)^2 \\
 &= \frac{1}{4} \cdot \frac{A_m^2}{2} = \frac{1}{8} A_m^2
 \end{aligned}$$

$$\therefore P_s = \frac{A_m^2}{4}$$

Total Power,  $P_t = P_c + P_s$

$$\begin{aligned}
 &= \frac{A_c^2}{2} + \frac{A_m^2}{4} \\
 &= \frac{A_c^2}{2} \left[ 1 + \frac{1}{2} \frac{A_m^2}{A_c^2} \right]
 \end{aligned}$$

$$P_t = P_c \left[ 1 + \frac{\mu^2}{2} \right] \quad \left[ \because \mu = \frac{A_m}{A_c} \right]$$

### Transmission Efficiency( $\eta$ ):

Transmission efficiency: It is The Percentage of The total power carried by The sidebands. It is denoted with  $\eta$

$$\begin{aligned}
 \eta &= \frac{P_s}{P_t} \times 100\% \\
 &= \frac{\frac{A_m^2}{4}}{\frac{A_c^2}{2} \left[ 1 + \frac{\mu^2}{2} \right]} \times 100\% \\
 &= \frac{\frac{A_m^2}{A_c^2}}{2 + \mu^2} \times 100 = \frac{\mu^2}{2 + \mu^2} \times 100\%
 \end{aligned}$$

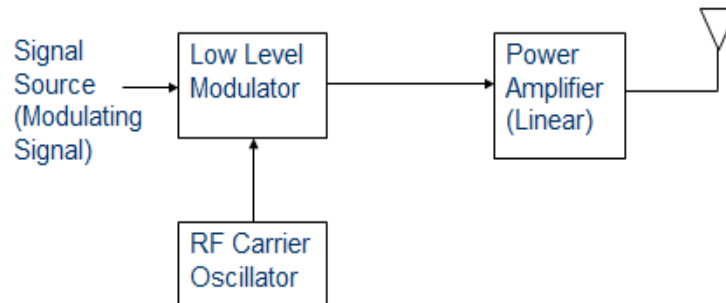
when  $\mu = 1 \Rightarrow \eta = 33.33\%$



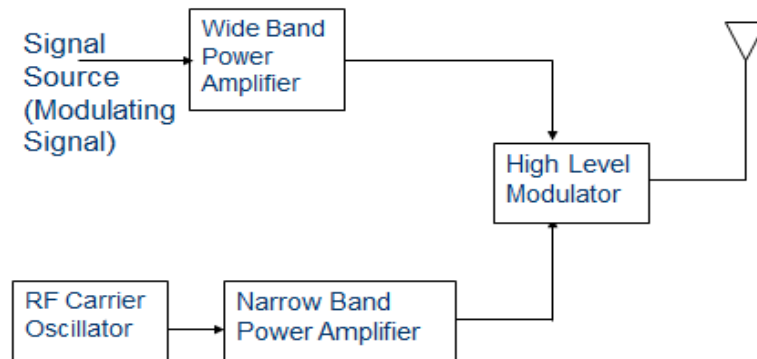
## Generation of AM waves:

The AM generation is classified as

**1. Low-level Modulation:** In this system modulation done at low power level. So very small power is associated with  $C(t)$  &  $m(t)$ . Due to this low output power of modulation power amplifiers are required. Example are Square-law and Switching modulator.



**2. High-level Modulation:** In this system modulation done at high power level. To do this both  $C(t)$  &  $m(t)$  must be at high power levels. So before modulation  $C(t)$  &  $m(t)$  are connected to power amplifiers.



**Note:** High level modulation produces best AM, but it requires an extremely high power modulator circuit which is expensive.

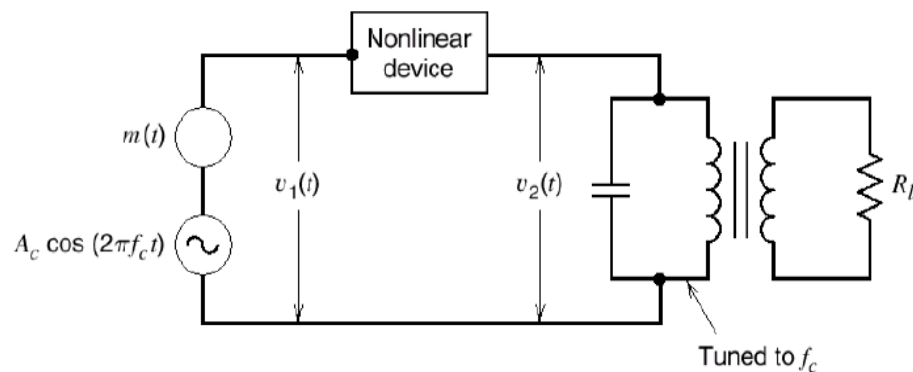
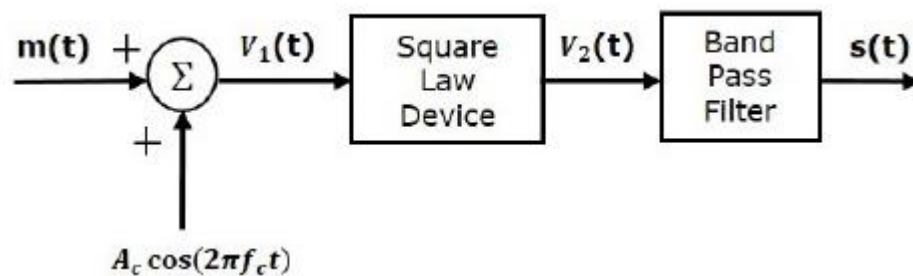
## Comparison of LOW & HIGH level modulations:

| S. No | Parameter        | LOW-LEVEL MODULATION                                                                    | HIGH-LEVEL MODULATION                                                          |
|-------|------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1     | Power level      | Modulation is carried out at low power level.                                           | Modulation is carried out at high power level.                                 |
| 2     | Amplifier stages | Need lesser amplifier stages.                                                           | Need more amplifier stages                                                     |
| 3     | Power efficiency | After modulation linear amplifiers can only be used. This gives lower power efficiency. | Non-linear amplifiers can also be used. This leads to higher power efficiency. |
| 4     | Power losses     | Power losses in amplifiers is higher, the cooling problem is severe.                    | Power losses is less, the cooling problem is not severe.                       |
| 5     | Applications     | Used as higher power broadcast transmitters.                                            | Used in TV transmitters                                                        |

## Square-law modulator:

A Square-law modulator requires three features: a means of summing the carrier and modulating waves, a nonlinear element, and a band pass filter for extracting the desired modulation products. Semi-conductor diodes and transistors are the most common nonlinear devices used for implementing square law modulators. The filtering requirement is usually satisfied by using a single or double tuned filters.

When a nonlinear element such as a diode is suitably biased and operated in a restricted portion of its characteristic curve, that is, the signal applied to the diode is relatively weak, we find that transfer characteristic of diode-load resistor combination can be represented closely by a square law :



$$V_0(t) = a_1 V_i(t) + a_2 V_i^2(t) \dots\dots\dots(i)$$

Where  $a_1, a_2$  are constants

Now, the input voltage  $V_i(t)$  is the sum of both carrier and message signals i.e.,

$$V_i(t) = A_c \cos 2\pi f_c t + m(t) \dots\dots\dots(ii)$$

Substitute equation (ii) in equation (i) we get

$$V_0(t) = a_1 A_c [1 + k_a m(t)] \cos 2\pi f_c t + a_1 m(t) + a_2 A_c^2 \cos^2 2\pi f_c t + a_2 m^2(t) \dots\dots\dots(iii)$$

Where  $k_a = 2a_2/a_1$

Now design the tuned filter /Band pass filter with center frequency  $f_c$  and pass band frequency width  $2W$ . We can remove the unwanted terms by passing this output voltage  $V_0(t)$  through the band pass filter and finally we will get required AM signal.

$$V_0(t) = a_1 A_c [1 + 2a_2/a_1 m(t)] \cos 2\pi f_c t$$

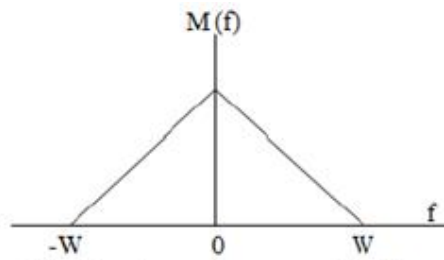


Fig. Spectrum of message signal

The Fourier transform of output voltage  $V_O(t)$  is given by

$$V_O(f) = a_1 A_c / 2 [\delta(f-f_c) + \delta(f+f_c)] + a_2 A_c [M(f-f_c) + M(f+f_c)]$$

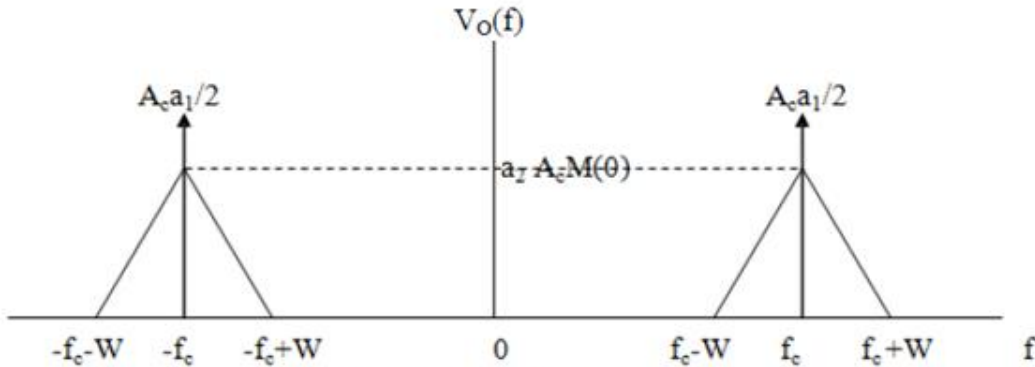


Fig: Spectrum of AM signal

### Switching modulator:

The total input for the diode at any instant is given by  $v_1 = c(t) + m(t)$

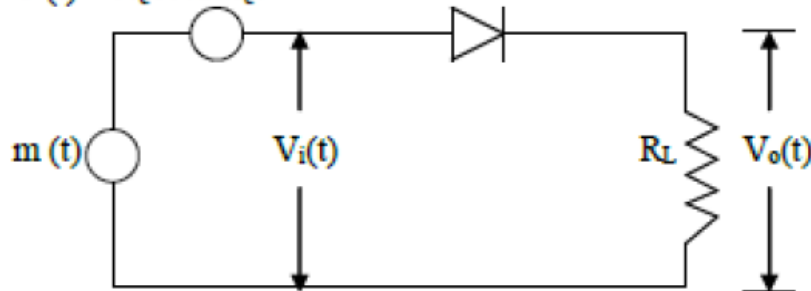
$$v_1 = A_c \cos 2\pi f_c t + m(t)$$

When the peak amplitude of  $c(t)$  is maintained more than that of information signal, the operation is assumed to be dependent on only  $c(t)$  irrespective of  $m(t)$ . When  $c(t)$  is positive,  $v_2 = v_1$  since the diode is forward biased. Similarly, when  $c(t)$  is negative,  $v_2 = 0$  since diode is reverse biased. Based upon above operation, switching response of the diode is periodic rectangular wave with an amplitude unity and is given by

$$p(t) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cos(2\pi f_c t (2n-1))$$

$$p(t) = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi f_c t) - \frac{2}{3\pi} \cos(6\pi f_c t) + \dots$$

$$C(t) = A_c \cos 2\pi f_c t$$



Therefore the diode response  $V_2$  is a product of switching response  $p(t)$  and input  $v_1$ .  $v_2 = v_1 * p(t)$

$$V_2 = [A_c \cos 2\pi f_c t + m(t)] \left[ \frac{1}{2} + \frac{2}{\pi} \cos 2\pi f_c t - \frac{2}{3\pi} \cos 6\pi f_c t + \dots \right]$$

Applying the Fourier Transform, we get

$$\begin{aligned} V_2(f) = & \frac{A_c}{4} [\delta(f - f_c) + \delta(f + f_c)] + \frac{M(f)}{2} + \frac{A_c}{\pi} \delta(f) - \frac{1}{3\pi} [M(f - 3f_c) + M(f + f_c)] \\ & + \frac{A_c}{2\pi} [\delta(f - 2f_c) + \delta(f + 2f_c)] + \frac{1}{\pi} [M(f - f_c) + M(f + f_c)] \\ & - \frac{A_c}{6\pi} [\delta(f - 4f_c) + \delta(f + 4f_c)] - \frac{A_c}{3\pi} [\delta(f - 2f_c) + \delta(f + 2f_c)] \end{aligned}$$

The diode output  $v_2$  consists of a dc component at  $f=0$ . Information signal ranging from 0 to  $w$  Hz and infinite number of frequency bands centered at  $f, 2f_c, 3f_c, 4f_c, \dots$

The required AM signal centered at  $f_c$  can be separated using band pass filter. The filter output is given by the equation

$$S(t) = \frac{A_c}{2} \left[ 1 + \frac{4}{\pi} \frac{m(t)}{A_c} \right] \cos 2\pi f_c t = S(t) = \frac{A_c}{2} \left[ 1 + \frac{4}{\pi} \frac{A_m}{A_c} \cos 2\pi f_m t \right] \cos 2\pi f_c t$$

Therefore modulation index,  $m = \frac{4}{\pi} \frac{A_m}{A_c}$

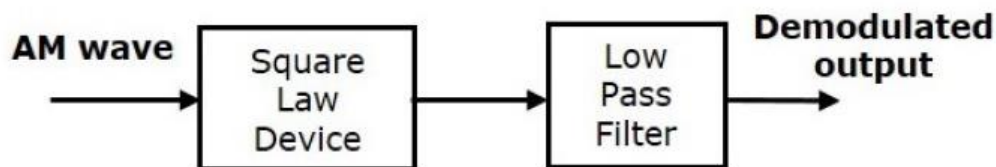
### Demodulation of AM waves:

There are two methods to demodulate AM signals. They are:

- Square-law detector
- Envelope detector

#### Square Law Demodulator

Square law demodulator is used to demodulate low level AM wave. Following is the block diagram of the square law demodulator.



This demodulator contains a square law device and low pass filter. The AM wave  $V_1(t)$  is applied as an input to this demodulator.

The standard form of AM wave is  $V_1(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t)$

We know that the mathematical relationship between the input and the output of square law device is

$$V_2(t) = k_1 V_1(t) + k_2 V_1^2(t) \quad \text{(Equation 1)}$$

Where,  $V_1(t)$  is the input of the square law device, which is nothing but the AM wave

$V_2(t)$  is the output of the square law device

$k_1$  and  $k_2$  are constants

Substitute  $V_1(t)$  in Equation 1



$$\begin{aligned}
V_2(t) &= k_1 (A_c [1 + k_a m(t)] \cos(2\pi f_c t)) + k_2 (A_c [1 + k_a m(t)] \cos(2\pi f_c t))^2 \\
\Rightarrow V_2(t) &= k_1 A_c \cos(2\pi f_c t) + k_1 A_c k_a m(t) \cos(2\pi f_c t) + k_2 A_c^2 [1 + K_a^2 m^2(t) + 2k_a m(t)] \left( \frac{1 + \cos(4\pi f_c t)}{2} \right) \\
\Rightarrow V_2(t) &= k_1 A_c \cos(2\pi f_c t) + k_1 A_c k_a m(t) \cos(2\pi f_c t) + \frac{K_2 A_c^2}{2} + \\
&\quad \frac{K_2 A_c^2}{2} \cos(4\pi f_c t) + \frac{k_2 A_c^2 k_a^2 m^2(t)}{2} + \frac{k_2 A_c^2 k_a^2 m^2(t)}{2} \cos(4\pi f_c t) + \\
&\quad k_2 A_c^2 k_a m(t) + k_2 A_c^2 k_a m(t) \cos(4\pi f_c t)
\end{aligned}$$

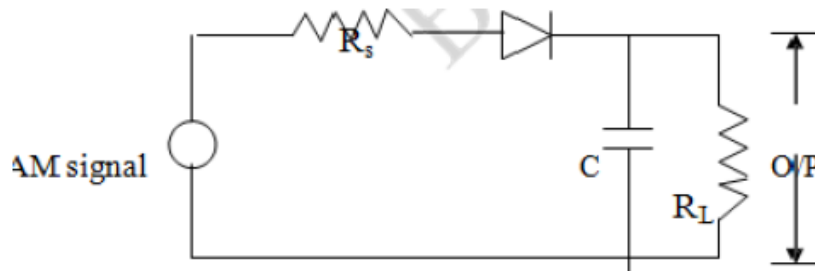
In the above equation, the term  $k_2 A_c^2 k_a m(t)$  is the scaled version of the message signal. It can be extracted by passing the above signal through a low pass filter and the DC component  $\frac{k_2 A_c^2}{2}$

can be eliminated with the help of a coupling capacitor.

### Envelope Detector:

Envelope detector is used to detect high level modulated levels, whereas square-law detector is used to detect low level modulated signals (i.e., below 1v). It is also based on the switching action or switching characteristics of a diode. It consists of a diode and a resistor-capacitor filter.

The operation of the envelope detector is as follows. On a positive half cycle of the input signal, the diode is forward biased and the capacitor C charges up rapidly to the peak value of the input signal. When the input signal falls below this value, the diode becomes reverse biased and the capacitor C discharges slowly through the load resistor  $R_L$ . The discharging process continues until the next positive half cycle. When the input signal becomes greater than the voltage across the capacitor, the diode conducts again and the process is repeated.



The charging time constant  $R_s C$  is very small when compared to the carrier period  $1/f_c$  i.e.,

$$R_s C \ll 1/f_c$$

Where  $R_s$  = internal resistance of the voltage source.

C = capacitor

$f_c$  = carrier frequency

i.e., the capacitor C charges rapidly to the peak value of the signal.

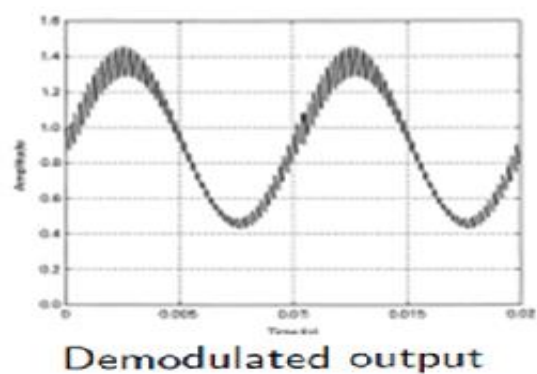
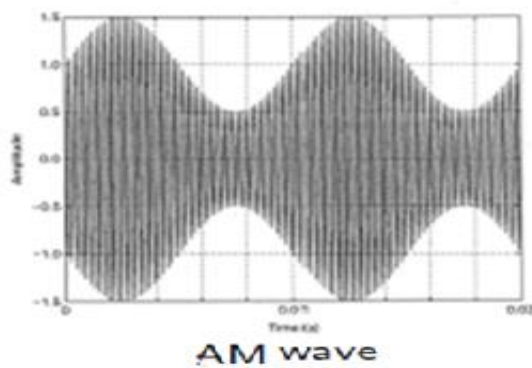
The discharging time constant  $R_L C$  is very large when compared to the charging time constant i.e.,

$$1/f_c \ll R_L C \ll 1/W$$

Where  $R_L$  = load resistance value

W = message signal bandwidth

i.e., the capacitor discharges slowly through the load resistor.



### Advantages:

- It is very simple to design
- It is inexpensive
- Efficiency is very high when compared to Square Law detector

### Disadvantage:

- Due to large time constant, some distortion occurs which is known as diagonal clipping i.e., selection of time constant is somewhat difficult

### Application:

- It is most commonly used in almost all commercial AM Radio receivers.

### NUMERICAL PROBLEMS ON AM

**Example-1** A modulating signal  $10 \sin(2\pi \times 10^3 t)$  is used to modulate a carrier signal  $20 \sin(2\pi \times 10^4 t)$ . Determine the modulation index, percentage modulation, frequencies of the sideband components and their amplitudes. What will be the bandwidth of the modulated signal?

**Solution:**

(i) The modulating signal  $v_m = 10 \sin(2\pi \times 10^3 t)$ .

Let us compare this with the following expression

$$v_m = V_m \sin(2\pi f_m t)$$

Then, we get  $V_m = 10 \text{ volt}, f_m = 1 \times 10^3 \text{ Hz} = 1 \text{ kHz}$

(ii) The carrier signal  $v_c = 20 \sin(2\pi \times 10^4 t)$

Comparing this with the expression  $v_c = V_c \sin(2\pi f_c t)$ , we obtain

$$V_c = 20 \text{ volt}, f_c = 1 \times 10^4 \text{ Hz} = 10 \text{ kHz}$$

(a) Modulation index and percentage modulation:

We know that  $m = \frac{V_m}{V_c} = \frac{10}{20} = 0.5$  and % modulation =  $0.5 \times 100 = 50 \%$

(b) Frequencies of sideband components :

(i) Upper sideband  $f_{\text{USB}} = f_c + f_m = (10 + 1) = 11 \text{ kHz}$

(ii) Lower sideband  $f_{\text{LSB}} = f_c - f_m = (10 - 1) = 9 \text{ kHz}$

(c) Amplitudes of sidebands:

The amplitudes of upper as well as the lower sideband are given by,

$$\text{Amplitude of each sideband} = \frac{mV_c}{2} = \frac{0.5 \times 20}{2} = 5 \text{ Volt}$$

(d) Bandwidth =  $2f_m = 2 \times 1 \text{ kHz} = 2 \text{ kHz}$

**Example-2. If the modulating frequency of a carrier wave varies between 700Hz and 7KHz, find it's bandwidth?**

**Explanation:** Modulating Frequency ( $f_m$ ) =  $V_{max} - V_{min}$ ,

where,  $V_{max}$  = Maximum Amplitude of an amplitude modulated,  $V_{min}$  = Minimum amplitude of an amplitude modulated,  $f_m = 7\text{KHz} - 700\text{Hz} = 6.3\text{KHz}$   
 Bandwidth =  $2f_m = 2 \times 6.3 = 12.6 \text{ KHz}$ .

**Example-3. A 400W carrier wave is modulated to a depth of 65%. Find the total power of modulated wave?**

**Explanation:** Total power,  $P_t = P_c (1 + \mu^2/2)$ , where  $P_c$  = Carrier Power = 400W  
 Given Modulation Index ( $\mu$ ) = 0.65,  
 So,  $P_t = 400 (1 + 0.65^2/2)$ . On solving it we get  $P_t = 484.5\text{W}$ .

**Example-4. A 1000 KHz carrier is modulated with 300 Hz, 8000 Hz and 2 KHz waves. Determine the frequencies whose chances of occurring in output is least?**

**Explanation:** Frequencies present in output are of the form  $f_c \pm f_m$ ,  $f_c \pm 2f_m$ ,  $f_c \pm 3f_m$ . And 1000 KHz is a multiple of none in the range. Whereas, rest options are one of the multiples in the range.

**Example-5. If average power of a transmitter is 4kW and maximum power is 20000KW. What is its duty cycle?**

**Explanation:** Duty cycle is the proportion of time during which a component, device, or system is operated. The duty cycle can be expressed as a ratio or as a percentage.

$$\text{Duty cycle} = \frac{\text{Average Power}}{\text{Peak Power}} = \frac{4}{20000} = 2 \times 10^{-4}$$

**Example-6 . A 400 watts carrier is modulated to a depth of 75 percent. Find the total power in the amplitude-modulated wave. Assume the modulating signal to be a sinusoidal one.**

**Solution :** We know that for a sinusoidal modulating signal, the total power is expressed as

$$P_t = P_c \left( 1 + \frac{m_a^2}{2} \right)$$

where

$P_t$  = total power or modulated power

$P_c$  = carrier power or unmodulated power

$m_a$  = modulation index

Given that,

$P_c = 400 \text{ watts}$

$m_a = 75 \text{ percent} = 0.75$

Therefore,

$$P_t = P_c \left( 1 + \frac{m_a^2}{2} \right) = 400 \left( 1 + \frac{0.75^2}{2} \right) = 512.5 \text{ watts}$$



The total modulation index  $m_t$  is given by

$$m_t = [m_1^2 + m_2^2]^{1/2} = [(0.6324)^2 + (0.5)^2]^{1/2} = 0.8$$

The new transmitted power is given by

$$P_t = P_c \left[ 1 + \frac{m_t^2}{2} \right] = 10 \text{ kW} \left[ 1 + \frac{(0.8)^2}{2} \right] = 13.2 \text{ kW}.$$

**Example-7** An AM broadcast radio transfer radiates 10 K watts of power if modulation percentage is 60. Calculate how much of this is the carrier power.

**Solution :** We know that the total power is expressed as

$$P_t = P_c \left( 1 + \frac{m_a^2}{2} \right)$$

where

$P_t$  = total power or modulated power

$P_c$  = carrier power or unmodulated power

$m_a$  = modulation index

Given that,

$$P_t = 10 \text{ K watts}$$

$$m_a = 60 \text{ percent} = 0.6$$

From equation (i), we get

$$P_c = \frac{P_t}{1 + \frac{m_a^2}{2}} = \frac{10}{1 + \frac{0.6^2}{2}} = \frac{10}{1.18} = 8.47 \text{ kW}$$

**Example-8** A certain transmitter radiates 10 kW with carrier unmodulated and 12 kW when the carrier is sinusoidally modulated. Calculate the modulation index. If another sinewave corresponding to 50% modulation is transmitted simultaneously, determine the total radiated power.

**Solution:** Given that  $P_c = 10 \text{ kW}$ ,  $P_t = 12 \text{ kW}$ ,  $m_2 = 0.5$

The modulation index  $m_1$  is given by,

$$\frac{P_t}{P_c} = 1 + \frac{m_1^2}{2}$$

or

$$\frac{12}{10} = \left[ 1 + \frac{m_1^2}{2} \right]$$

or

$$m_1 = 0.6324$$

**Example-9**

An AM transmitter radiates 9 kW of power when the carrier is unmodulated and 10.125 kW when the carrier is sinusoidally modulated. Find the modulation index, percentage of modulation. Now, if another sine wave, corresponding to 40 percent modulation is transmitted simultaneously, then calculate the total radiated power. (Very Important)

**Solution :** (i) We know that for a single-tone sinusoidal amplitude-modulation the total power is expressed as

$$P_t = P_c \left( 1 + \frac{m_a^2}{2} \right)$$

where

$P_t$  = modulated or total power

$P_c$  = unmodulated or carrier power

$m_a$  = modulation index

Given that,

$$P_t = 10.125 \text{ kW}$$

and

$$P_c = 9 \text{ kW}$$

Using equation (i), we get



$$1 + \frac{m_a^2}{2} = \frac{P_t}{P_c}$$

or  $\frac{m_a^2}{2} = \frac{P_t}{P_c} - 1 = \frac{10.125}{9} - 1$

or  $\frac{m_a^2}{2} = 1.125 - 1 = 0.125$

or  $m_a^2 = 0.125 \times 2 = 0.250 = 0.50$

(ii) We know that in case of modulation by two sinusoidal waves, the total modulation index  $m_t$  is expressed as

$$m_t = \sqrt{m_1^2 + m_2^2}$$

Let  $m_1 = m_a = 0.5$

Given that  $m_2 = 0.4$

Therefore,  $m_t = \sqrt{(0.5)^2 + (0.4)^2} = \sqrt{0.25 + 0.16}$

or  $m_t = \sqrt{0.41} = 0.64$

The total radiated power in this case will be

$$P_t = P_c \left( 1 + \frac{m_t^2}{2} \right) = 9 \times \left( 1 + \frac{0.64^2}{2} \right)$$

$$P_t = 9 (1 + 0.205) = 10.84 \text{ kW}$$

**Example-10** The carrier amplitude after AM varies between 4 volts and 1 volt. Calculate depth of modulation.

**Solution:** We know that we can calculate the value of modulation index with the help of the following expression :

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

Substituting  $V_{\max} = 4 \text{ Volt}$  and  $V_{\min} = 1 \text{ Volt}$ , we get,

$$m = \frac{4 - 1}{4 + 1} = \frac{3}{5} = 0.6 \text{ or } 60\%$$

**Example-11** Prove that in amplitude modulation, maximum average power transmitted by an antenna is 1.5 times the carrier power.

**Solution:** Let  $P_c$  = Unmodulated carrier power

$P_t$  = Transmitted AM power,  $m$  = Modulation index

We know that  $P_t = P_c \left[ 1 + \frac{m^2}{2} \right]$

The maximum value of  $m$  without introducing distortion in the modulated wave is  $m = 1$ . Substituting this value in equation (i), we obtain

$$P_t(\max) = \left[ 1 + \frac{1}{2} \right] = 1.5 P_c$$

**Example-12** In an AM system, the modulating signal is sinusoidal with frequency  $f_m$  Hz. If 80% modulation is used, then find the ratio of total side-band power in the modulation signal, to the total power.

**Solution :** The AM signal is expressed as

$$\begin{aligned} x_{AM}(t) &= A \cos \omega_c t + mA \cos \omega_m t \cos \omega_c t \\ &= A \cos \omega_c t + \frac{mA}{2} [\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t] \end{aligned}$$

$$P_c \text{ (carrier power)} = \frac{1}{2} A^2$$

$$P_s \text{ (sideband power)} = \frac{1}{2} \left[ \left( \frac{1}{2} mA \right)^2 + \left( \frac{1}{2} mA \right)^2 \right] = \frac{1}{4} m^2 A^2$$

Total power is given by

$$P_t = P_c + P_s = \frac{1}{2} \left( 1 + \frac{m^2}{2} \right) A^2$$

$$\begin{aligned} \text{Ratio} &= \frac{\text{Sideband power}}{\text{Total power}} = \frac{1/4 m^2 A^2}{1/4 (2 + m^2) A^2} = \frac{m^2}{2 + m^2} \\ &= \frac{(0.80)^2}{2 + (0.80)^2} = \frac{0.64}{2.64} = 0.242. \end{aligned}$$

**Example-13** Determine the percentage power saving when the carrier wave and one of the side bands are suppressed in an AM wave modulated to a depth of (i) 100% and (ii) 50%.

**Solution:** (i) Total power in AM wave is given by

$$P_t = P_c \left( 1 + \frac{m^2}{2} \right)$$

Given that modulation index  $m = 100\% = 1.0$

Equation (i) can be written as

$$P_t = P_c \left( 1 + \frac{(1)^2}{2} \right) = 1.5 P_c$$

The power in one side band is given by

$$P_{SB} = P_c \frac{m^2}{4} = P_c \frac{(1)^2}{4} = \frac{P_c}{4} = 0.25 P_c$$

$$\text{Power saving} = \frac{P_t - P_{SB}}{P_t} = \frac{1.5 P_c - 0.25 P_c}{1.5 P_c} = \frac{1.25}{1.50} = \frac{5}{6} = 0.833 = 83.3\%$$

(ii) Given that  $m = 50\% = 0.50$  and  $P_t = P_c \left( 1 + \frac{m^2}{2} \right)$

Putting the values of  $m$ , we have

$$P_t = P_c \left( 1 + \frac{(0.5)^2}{2} \right) = 1.125 P_c$$

$$P_{SB} = P_c \frac{m^2}{4} = P_c \frac{(0.5)^2}{4} = 0.0625 P_c$$

Hence,  $\text{power saving} = \frac{P_t - P_{SB}}{P_t} = \frac{1.125 P_c - 0.0625 P_c}{1.125 P_c} = 0.944 = 94.4\%$